

Alireza Montajab

Electrical Engineering | Digital Electronics

Personal Information

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Education

2023–2027 (expected) **Bachelor of Science in Electrical Engineering**, *Shahid Beheshti University*, Tehran, Iran
Specialisation: Digital Electronics (from February 2026)
GPA: 16.50/20 | Major GPA: 16.00/20
QS World University Rankings 2026: #741–750 worldwide, #214 in Asia
Relevant coursework: Digital Systems I & II, Logic Circuits, Microprocessors, Electronics I–III, Signals & Systems, Linear Control Systems, Fundamentals of Mechatronics

Research Interests

Embedded and real-time systems, closed-loop control on constrained hardware, sensor interfacing and vision-based feedback, digital system and datapath design.

Experiences

Work Experience

2026–present **Embedded Hardware Engineer**, *Tensor Intelligent Machines*, Tehran, Iran
Firmware and PCB design for robotic systems at Sharif University Science & Technology Park.
2025–2026 **Engineering Intern**, *Tensor Intelligent Machines*, Tehran, Iran
Embedded firmware and PCB design; converted to a full-time engineering role.

Teaching Experience

Altium Designer PCB Design Workshop, *Instructor*, Mechatronics Student Association, Shahid Beheshti University
Electrical Circuits, *Teaching Assistant*, Shahid Beheshti University

Projects

Industry Projects

2026 **Handheld Barcode Terminal**, *Tensor Intelligent Machines*
ESP32-S3 firmware using the USB Host HID stack and FreeRTOS, with a custom carrier PCB.
2025 **RFID Reader Board**, *Tensor Intelligent Machines*
Surface-mount reader board designed in Altium Designer, with a fully sourced bill of materials.

- 2025 **Ultrasonic Ranging Module**, *Tensor Intelligent Machines*
Microsecond-resolution distance measurement on an STM32G0 using hardware timer capture.

Academic Projects

- Spring 2026 **Magnetic Levitation System with Visual Feedback**, *Fundamentals of Mechatronics*, Supervisor: Dr. Nouri-Manzar
Stabilised an inherently unstable maglev system using an ESP32-CAM as the position sensor and an on-board PID controller.
- Fall 2025 **Water Level and Temperature Control System**, *Digital Systems II*, Supervisor: Dr. Asharioun
Bare-metal STM32 firmware with two control loops, timer-capture ultrasonic sensing and an LCD/keypad menu. Led a team of three.
- Spring 2025 **8-Bit Arithmetic Datapath**, *Digital Systems I*, Supervisor: Dr. Farhad Pouladi
Designed an 8-bit datapath from discrete 74LS logic: adders, multiplexers, a decoder and shift registers.
- Fall 2025 **Wi-Fi Controlled Smart Lighting**, *Digital Systems II*
ESP32 web server with REST endpoints controlling four lighting channels.
- Spring 2025 **H-Bridge Motor Driver PCB**, *Electronics II*
Four-transistor bidirectional DC motor driver designed in Altium Designer.
- Fall 2025 **Transmission Line Parameter Calculator**, *Power System Analysis I*
MATLAB tool computing inductance and capacitance of bundled three-phase lines.
- Spring 2024 **Op-Amp RLC Network Analysis**, *Electrical Circuits I*
Closed-form transient and steady-state analysis, verified in LTspice.

Technical Skills

- Embedded C, C++, Embedded C, STM32 (HAL, CubeMX), ESP32 / ESP32-S3 (ESP-IDF), FreeRTOS, ARM Cortex-M
- Interfaces USB HID, UART, I²C, SPI, PWM, ADC, DMA, timers, input capture, interrupts
- Hardware Altium Designer, SMD PCB design, BOM sourcing, MOSFET power stages, motor drivers, sensor interfacing
- Digital Design 74-series logic, finite state machines, datapath design
- Control PID design and tuning, closed-loop feedback, real-time image processing
- Tools Proteus, LTspice, MATLAB / Simulink, oscilloscope, logic analyser, Git, Linux, LaTeX

Awards and Honors

- Awarded a full tuition scholarship at Shahid Beheshti University (2023–2027), granted on national entrance examination rank.
- Ranked within the top 1% of the Iranian University Entrance Exam (2023).

Languages

- English IELTS Academic 7.0
- Persian Native